

CVIS Hands-on Training on Development of Machine Learning and Deep Learning Models Summer 2025

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Overview:

- Image preprocessing and augmentation
- CNN architectures:
 - VGG
 - ResNet
 - Transfer learning and fine-tuning
 - U-Nets
 - Generative Adversarial Networks (GAN)



01. Image preprocessing and augmentation

- Image augmentation is a technique used to artificially increase the diversity of a training dataset by creating modified versions of existing images.
 - Model reproducibility is increased.
 - Training time is decreased.
 - Training costs are decreased.
- Read about them here:
 - <https://docs.roboflow.com/datasets/dataset-versions/image-augmentation>
- Other augmentation can lead to creation of synthetic image:
 - <https://github.com/UjjwalSaxena/Automold--Road-Augmentation-Library>



02. Network architectures

- ❏ Classification models
- ❏ Generative models

Classification Demo:

<https://colab.research.google.com/drive/1zlqai189eGw7513DW1cliXPB3eSl-sTl>

Autoencoders Demo:

https://colab.research.google.com/drive/1YehVcHnoy8QLS_Fqj9CE8OzEEYHeDjdO

Vgg 16



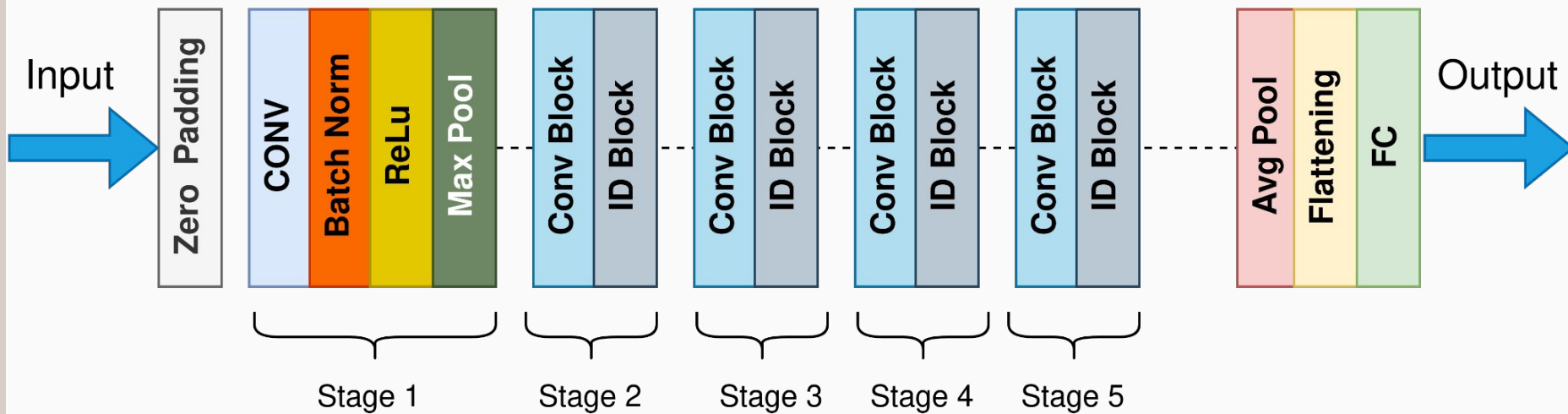
VGG - 16



<https://medium.com/@teja.nsrk97/a-beginners-guide-to-vgg16-the-visual-detective-of-image-classification-b45cd81935d7>

ResNet50

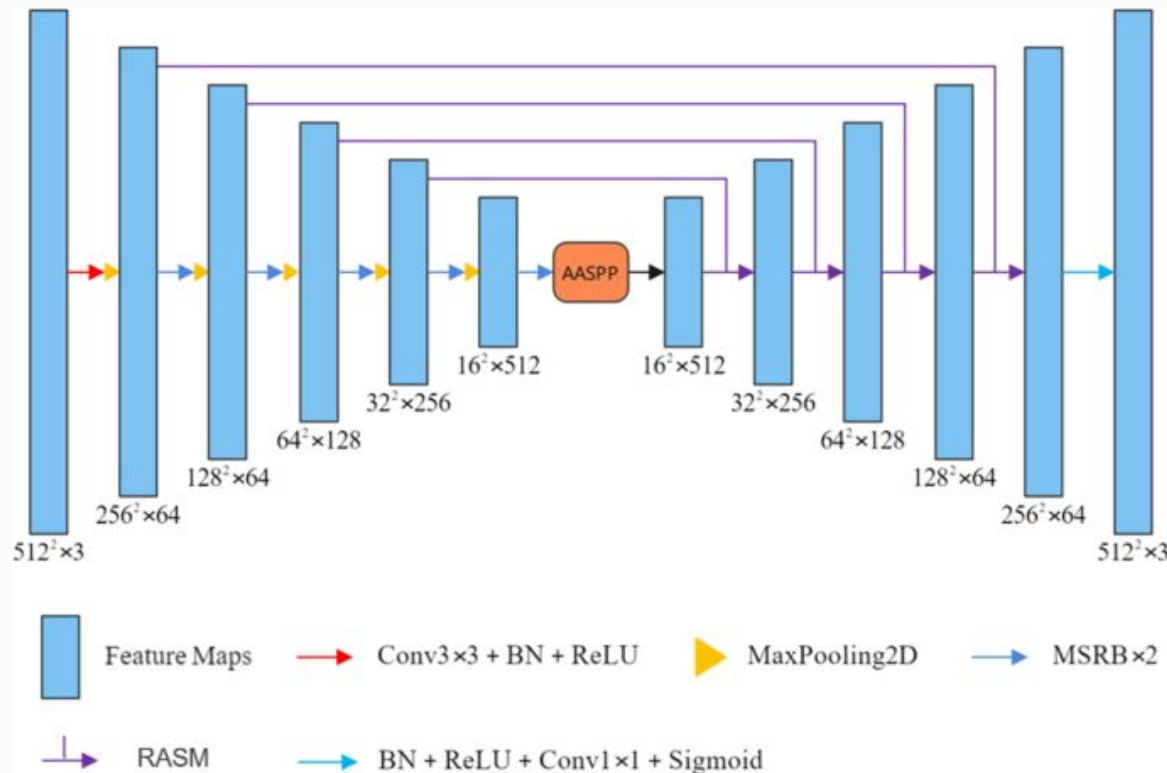
ResNet50 Model Architecture



<https://medium.com/@nitishkundu1993/exploring-resnet50-an-in-depth-look-at-the-model-architecture-and-code-implementation-d8d8fa67e46f>

Autoencoders & U-Nets

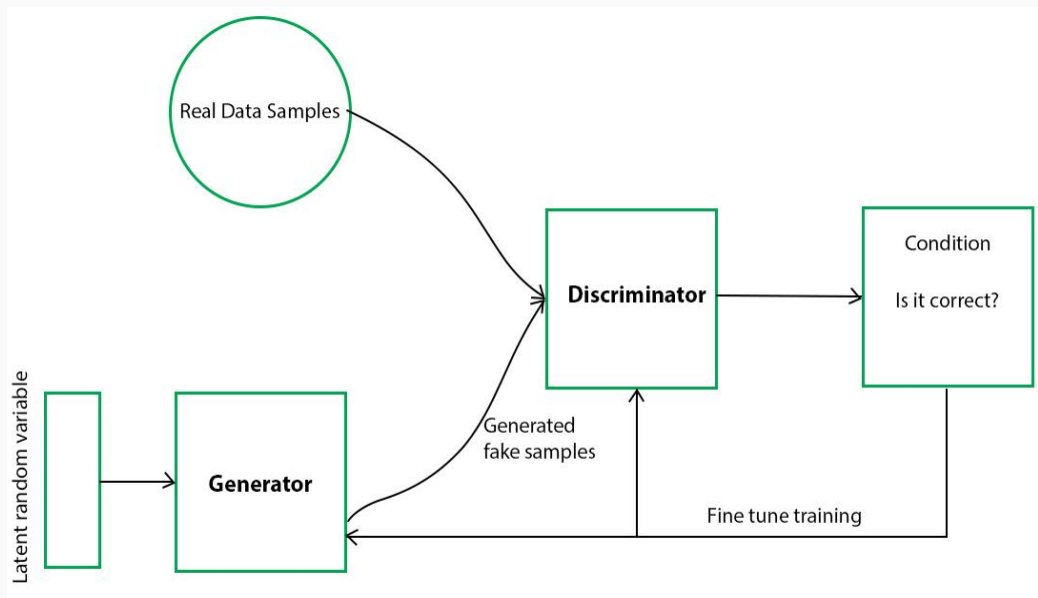
- Encoding Path
- Decoding Path
- Skip connections



<https://medium.com/@fernandopalominocobo/mastering-u-net-a-step-by-step-guide-to-segmentation-from-scratch-with-pytorch-6a17c5916114>

GANs

- Generator. Model that is used to generate new plausible examples from the problem domain.
- Discriminator. Model that is used to classify examples as real (from the domain) or fake (generated).



<https://machinelearningmastery.com/what-are-generative-adversarial-networks-gans/>



Helpful Youtube Links

Here is a doc file containing all the links I share with my CSE 463 students-

- https://docs.google.com/spreadsheets/d/e/2PACX-1vSa1z9ODwH3s74PZo7QIPGn8kPfy0cyf75Su98ZL54sqYiU8X9DsII_I6pT8SUhHiJ6FZ8TgsBQG8H_/pubhtml?gid=1740133678&single=true